Artifact Technical Specifications

This document summarizes the technical infrastructure, experimental configuration, conversational and call-management settings, and data collection and monitoring procedures for the AI voice agents proposed for use in the field experiment.

1. *AI Infrastructure and Performance*

This subsection outlines the technologies used to support real-time AI voice conversations, including speech recognition, response generation, voice synthesis, and data infrastructure. It also reports key performance metrics and estimated operating costs.

- **AI Voice Agent Platform**: Vapi.ai, which coordinates real-time conversations between participants and the AI agents.
- **Speech Transcription:** Deepgram's Nova-2 model converts participants' speech into text, with a reported latency of 290 milliseconds, a word error rate of 3.3%, and a cost of approximately $0.01 per minute.
- **Conversational AI:** OpenAI's GPT-4o Mini Cluster model generates the agents' conversational responses, with an estimated latency of 560 milliseconds and a cost of $0.01 per minute.
- **Voice Generation:** Vapi generates the agents' spoken responses, with a latency of approximately 430 milliseconds and a cost of $0.02 per minute. The platform reports a voice human-likeness score of 92/100.
- **Overall Performance:** Combined reported processing latency is approximately 1.28 seconds, with an estimated total operating cost of $0.10–$0.15 per call minute.
- **Calling and Data Infrastructure:** Twilio handles telephone connectivity, while Amazon Web Services (AWS) hosts the cloud infrastructure and database.
- **Workflow Automation:** Make.com coordinates automated calling workflows through a custom webhook (HTTP POST to /call, with an HTTP 200 success response).

2. *Conversational Timing and Turn-Taking*

This subsection details how the AI agents manage speaking, listening, and interruptions during calls. These settings are designed to support natural conversational exchanges while allowing participants to speak without unnecessary interruptions.

- **Speaking Delay:** The agent is configured to begin speaking after a 0.2-second delay.
- **Speech Detection:** Vapi identifies when participants begin and finish speaking to support natural conversational turn-taking.
- **Interruption Handling:** The agent is configured to stop speaking when a participant interrupts (configured thresholds: one word and 0.2 seconds of participant speech).
- **Response After Interruption:** The configured restart delay following an interruption is 0 seconds.

3. *Voicemail Detection and Call Limits*

This subsection describes how the system handles unanswered calls, voicemail detection, and call duration. These settings establish consistent procedures for managing calls across participants.

- **Voicemail Detection:** Vapi checks for voicemail beginning 5 seconds after connection, with subsequent checks every 5 seconds for up to five attempts. The additional wait before leaving a voicemail is configured to 0 seconds.
- **Inactivity Timeout:** Calls automatically terminate after 30 seconds of inactivity.
- **Maximum Call Duration:** Each call is limited to 300 seconds (5 minutes).

4. *Experimental Configuration*

This subsection describes how the AI agents will be deployed in the field experiment, including random assignment, call frequency, and conversational structure. It also outlines planned procedures for ensuring consistency across experimental conditions.

- **Random Assignment:** Participants are randomly assigned to one of four experimental conditions with equal probability (25% per condition).
- **Call Schedule:** Participants receive one AI check-in call per week over a four-week period.
- **Check-In Structure:** Calls include six core check-in questions, with additional tailored follow-up questions permitted in the agentic condition.
- **Implementation Fidelity:** Automated testing is planned to verify that agents consistently follow their assigned conversational protocols.
- **Recovery Assessment:** Integration of the 10-item Brief Assessment of Recovery Capital (BARC-10) is under consideration.

5. *Data Collection and Monitoring*

This subsection outlines how information from AI check-ins is organized and categorized through a centralized dashboard (note: reviewer login instructions are located in a separate document titled "Artifact Validation Instrument" and in Appendix E of the online supplement). The system is designed to capture participant data and monitor participant engagement, recovery-related challenges, and potential risk indicators.

- **Data Dashboard:** A centralized dashboard integrates with the OpenAI API to organize check-in information for monitoring and analysis.
- **Check-In Status Categories:** The dashboard classifies outcomes as *Doing Well*, *Struggling*, *At Risk*, or *No Response*.
- **Risk Flag Categories:** Separate indicators classify potential concerns as *Normal*, *Elevated Risk*, or *Urgent*.